MTH645 - Fuzzy Logic and Applications (Quiz No. 4) **Quiz Start Time: 06:52 PM, 09 January 2025**

Question # 2 of 10 (Start time: 06:52:57 PM, 09 January 2025) **Total Marks:** 1

If two fuzzy sets are identical, what is the distance between them?

Select the correct option

Reload Math Equations

- ○ Zero
- ○ One
- ○ It cannot be determined
- ○ Infinity

Click to Save Answer & Move to Next Question

sec(s)

MTH645 - Fuzzy Logic and Applications (Quiz No. 4) **Quiz Start Time: 06:52 PM, 09 January 2025**

Question # 4 of 10 (Start time: 06:54:53 PM, 09 January 2025) **Total Marks:** 1

For two Intuitionistic Fuzzy Sets A and B, their intersection is defined by which of the following operations on their membership and non-membership functions?

Select the correct option

Reload Math Equations

○	$\mu_A(x)\mu_B(x)\ and\ \nu_A(x)\nu_B(x)$
○	$\min\left((\mu_A(x),\mu_B(x))\ and\ \max\left(\nu_A(x),\nu_B(x)\right)\right)$
○	$\mu_A(x)+\mu_B(x)\ and\ \nu_A(x)+\nu_B(x)$
○	$\max\left((\mu_A(x),\mu_B(x))\ and\ \min\left(\nu_A(x),\nu_B(x)\right)\right)$

Click to Save Answer & Move to Next Question

MTH645 - Fuzzy Logic and Applications (Quiz No. 4) **Quiz Start Time: 06:52 PM, 09 January 2025**

Question # 5 of 10 (Start time: 06:55:36 PM, 09 January 2025) **Total Marks:** 1

For an Intuitionistic Fuzzy Set, if $\mu_A(x) = 0.8$ and $\nu_A(x) = 0.15$, what can we say about the hesitancy degree?

Select the correct option

Reload Math Equations

- It is equal to 0.85
- It is equal to 0.65
- It cannot be determined from the given information
- It is equal to 0.05

Click to Save Answer & Move to Next Question

sec(s)

MTH645 - Fuzzy Logic and Applications (Quiz No. 4) **Quiz Start Time: 06:52 PM, 09 January 2025**

Question # 6 of 10 (Start time: 06:56:05 PM, 09 January 2025) **Total Marks:** 1

Pick up the intuitionistic fuzzy set.

Select the correct option

Reload Math Equations

○	{(x,(0.5,0.6))}
○	{(x,(0.7))}
○	{(x,(0.6,0.3))}
○	{(x,(0.3,0.6,0.3))}

Click to Save Answer & Move to Next Question

MTH645 - Fuzzy Logic and Applications (Quiz No. 4) **Quiz Start Time: 06:52 PM, 09 January 2025**

Question # 1 of 10 (Start time: 06:52:19 PM, 09 January 2025) **Total Marks:** 1

If A(x,0.3) and B(x,0.35) are two fuzzy sets, then max(A, B)=------

Select the correct option

Reload Math Equations

- 0
- 0.35
- 0.3
- 0.5

Click to Save Answer & Move to Next Question

MTH645 - Fuzzy Logic and Applications (Quiz No. 4)

Quiz Start Time: 06:52 PM, 09 January 2025

Question # 7 of 10 (Start time: 07:02:07 PM, 09 January 2025)

Total Marks:

If the membership degree of IFS is 0.9 then the non-membership degree of it is-------

Select the correct option

Reload Math Equations

- 0
- 0.9
- 0.1
- 0.5

Click to Save Answer & Move to Next Question

MTH645 - Fuzzy Logic and Applications (Quiz No. 4) **Quiz Start Time: 06:52 PM, 09 January 2025**

Question # 8 of 10 (Start time: 07:02:49 PM, 09 January 2025) **Total Marks:** 1

Let $X=\{x_1, x_2, ..., x_n\}$ be a universal set and A, B are two fuzzy sets on X, the distance d(A, B) satisfy the condition:

Select the correct option

Reload Math Equations

○	$1 \leq d(A, B)$
○	$0 \leq d(A, B) \leq 1$
○	$-1 \leq d(A, B) \leq 0$
○	$d(A, B) \leq 0$

Click to Save Answer & Move to Next Question

MTH645 - Fuzzy Logic and Applications (Quiz No. 4) **Quiz Start Time: 06:52 PM, 09 January 2025**

Question # 9 of 10 (Start time: 07:03:19 PM, 09 January 2025) **Total Marks:** 1

If the membership degree of IFS is 0.2 then the non-membership degree of it is-------

Select the correct option

Reload Math Equations

- 0.8
- 0
- 0.2
- 0.4

Click to Save Answer & Move to Next Question

sec(s)

MTH645 - Fuzzy Logic and Applications (Quiz No. 4) **Quiz Start Time: 06:52 PM, 09 January 2025**

Question # 10 of 10 (Start time: 07:03:54 PM, 09 January 2025) **Total Marks:** 1

What is the complement of an Intuitionistic Fuzzy Set A with membership function $\mu_A(x)$ and non-membership function $\nu_A(x)$?

Select the correct option

Reload Math Equations

- $(\nu_A(x), 1 - \mu_A(x))$
- $(\nu_A(x), \mu_A(x))$
- $(1 - \mu_A(x)), 1 - \nu_A(x))$
- $(1 - \nu_A(x), \mu_A(x))$

Click to Save Answer & Move to Next Question